From AI Strategy to University Rules in the GCC

***Abstract*—How can a reader check whether national artificial-intelligence (AI) ambitions are visible in rules that university staff and students can use? We present the Four-Layer Public-Evidence Audit (4L-PEA), an Arabic–English document protocol of the six Gulf Cooperation Council (GCC) states. It separates national direction (L1), cross-sector operational governance (L2), higher-education (HE) controls (L3), and institution-level rules (L4). Each layer is classified as explicit (E), partial (P), or unlocated (U) under declared evidence gates. At the fixed 23 July 2026 cutoff, the 36-record official-source frame identifies L4 evidence in all six states, but qualifying L3 evidence in only one. A rule-based panel of 57 institutions contains 20 E, 12 P, and 25 U cases. A broader reading of the UAE HE evaluation framework raises L3 to two states, leaving four documentary L3–L4 mismatches. To make the method reusable, the paper supplies an auditable four-step workflow, a worked UAE example, decision rules, a 57-institution appendix, and executable count checks. A secondary interval-valued fuzzy representation displays the limits of partial evidence; it does not validate missing documents or coder judgments. Public visibility is not proof of legal transmission or implementation.**

***Index Terms*—AI governance, GCC, higher education, bilingual document audit, public traceability, evidence admissibility, policy reuse.**

I. Why This Audit Is Needed

An instructor needs to know whether AI use is permitted in an assignment, what must be disclosed, and who verifies a student's work. A national strategy can express ambition without answering those questions. The practical research question is therefore narrower than "Which country has the best AI governance?": *Which official documents can a reader actually locate and check at each governance level?*

Prior studies have compared GCC national strategies [1] and university AI policies across 40 universities in six global regions [2], 116 US research institutions [3], and 343 institutions in five countries [4]. Work on assessment [5], university policy development [6], and systematic or institution-level governance [7], [8], [9], [10], [11] illustrates substantial policy variety. Multi-level governance and policy translation are distinct concepts [12], [13]. Our contribution is the Four-Layer Public-Evidence Audit (4L-PEA), a *source-to-claim procedure*: it shows how to record the evidence for four levels without assuming that a policy at one level caused, authorized, or was implemented by another.

A reader example. Suppose a ministry advertises AI innovation, while a university publishes a formal rule specifying allowed assistance in assessments. The ministry announcement can document ambition but cannot tell the student what that university permits. Our audit asks two different questions: *does an HE-specific, operative instrument exist at the sector level?* and *does a university-issued rule exist at a named institution?* It records direct textual evidence for each answer. It never assumes that one document caused the other or that staff comply with the document simply because it is online.

The paper answers three questions: (1) Which layers contain a qualifying public instrument? (2) Does the difference between sector-level HE evidence and institutional rules persist under stated evidence and sampling changes? (3) What minimum record would enable another researcher to check each classification? The accompanying material gives that reader a workflow, not a new country score.

II. The Four-Layer Decision Rule

A. What E, P and U mean

For each GCC jurisdiction j and layer ℓ, we record one nominal state $L_{\ell j} \in \{\mathsf{E}, \mathsf{P}, \mathsf{U}\}$. **E (explicit)** means a dated, inspectable official instrument meets the layer's authority, scope and substantive-content gate. **P (partial)** means relevant official evidence exists but misses at least one required element (for example, a draft, sector-mismatched guidance, or a single-course rule). **U (unlocated)** means a qualifying public instrument was not found through the recorded search. U is *not* evidence that an internal policy does not exist.

TABLE I
A short rulebook for deciding whether evidence is explicit

Level	To record E, the official document must show:
L1	Adopted or authoritative *national AI direction*.
L2	An inspectable *cross-sector operational* AI control, standard, regulation or guideline.
L3	*HE-applicable sector guidance* that actually specifies AI-use controls (not only readiness metrics).
L4	*At least one university/college rule* with institutional authority and applicable AI-use provisions.

The L3/L4 thresholds answer different questions. L3 concerns the existence of an HE-sector instrument; L4 asks only whether *one institution* has an applicable rule. L4=E does not mean that most universities have a policy. Conversely, the absence of a qualifying public L3 document does not invalidate an L4 rule. The mismatch is a comparison of two different, explicitly stated *document-availability propositions*, not an estimate of sector compliance.

B. Availability, documentary links and real use are separate

Three kinds of evidence should never be conflated: *availability* is the public text of a qualifying policy at L1–L4; an *explicit link* is a clause in one named instrument citing, delegating to, or applying another named instrument; and *implementation* is observed use, monitoring, enforcement or practice. **Only availability was coded in this study.** A four-layer E/E/E/E row proves neither a directed policy chain nor implementation. A future link study must record instrument-to-instrument citations in a separate edge table and require the controlling passage.

III. How to Apply the Method

Figure 1 is a reusable procedure for a country, education system or group of institutions. **Step 1: freeze** the decision question, document cutoff, official institutional frame, language scope and exact search terms *before* seeing the outcome. **Step 2: locate** government, regulator, ministry and university texts, using the original document rather than a search snippet; record issuer, title, version, date, language, URL and pinpoint clause. **Step 3: code** each layer using Table I; keep any explicit cross-document citation in another field and never fill gaps by inference. **Step 4: report** all E/P/U states, missing evidence, sensitivity to disputed boundaries and raw source records. The reusable blank log and bilingual query templates accompany the PDF; they are *prospective templates*, not a claim that every original search string was historically retained.

For a U closure, search both Arabic and English on the issuing institution's domain, its policy library, teaching/assessment pages, student handbook and integrity rules, recording the completed surfaces and date. A ministry announcement that guidance is under development can be P; it is not automatically an enacted HE control. If Arabic and English editions disagree, record the scope, controlling-language clause and version authority and flag the classification for adjudication, rather than silently choosing a convenient translation. This is especially relevant where an English text is identified as a translation of an Arabic original [25].

A. Worked example: how the same country can have E, E, P, E

The UAE provides a concise worked classification under the fixed cutoff:

Layer	Located official instrument / decision reason	State
L1	National AI strategy: national direction	E
L2	UAE AI charter: cross-sector guidance	E
L3	HE OBEF v11.5: evaluates AI readiness, not operative use controls [15]	P
L4	UAEU student GenAI use policy: institutional rule	E

The profile is **E/E/P/E**. It says what is publicly *documentable*, not that the university policy was caused by the evaluation framework. Under an explicitly broader *evaluation-framework* threshold, UAE L3 changes to E; this decision is reported separately as sensitivity, never as a silent recode. The classification was not independently adjudicated by two coders.

IV. The GCC Evidence Frame

The fixed source date is **23 July 2026**. The original 36-record register comprises seven L1, eight L2, five L3, eleven L4, and five GCC-regional records. Nineteen records have an Arabic and English location, fifteen an English location only, and two an Arabic location only. The register's document titles, issuer, date, locator and classification rationale are available online, but the full set of verbatim excerpts and independent archival hashes has *not* been reverified for this revision. Source recovery is not retrospectively presented as a fresh complete search.

The jurisdictional L4 finding requires only one qualifying university. We added a *separate, nonrandom corroboration panel*: officially recognized or licensed civilian HEIs, grouped by G1 public/general, G2 private/non-government and G3 public applied/specialized. The declared oldest-four-with-ties rule was applied inside each country–stratum cell, retaining smaller cells and excluding author-affiliated institutions before AI-policy search. The founding year is a selection variable, not a maturity score.

Two eligibility corrections changed the selected members without changing that rule: a historical Bahrain regulator source placed Kingdom University in its eligible cell [16], [17]; UAE accreditation evidence established Emirates Institute for Banking and Financial Studies (EIBFS) as an eligible older private institution, displacing two later cases [18], [19]. The corrected panel is **57 institutions**. Importantly, the complete original pre-outcome eligible universe was not retained; thus the corrected membership can be audited for known cases, but this is *not* a claim of a reconstructed, fully frozen population frame.

V. Results: What the Documents Show

Table II makes the main result visible without a color-coded country score. National AI direction is E in four states, operational governance in five, HE-sector controls in one and an institution-level rule in all six. In *five of six states*, a visible L4 rule coexists with an L3 state other than E. Define the descriptive count

$$D = \#\{j : L_{4j} = \mathsf{E} \text{ and } L_{3j} \neq \mathsf{E}\} = 5. \quad (1)$$

A higher level's partial or unlocated public documentation cannot be interpreted as proof of a broken regulation or of institutional non-compliance. The 57-case panel contains **20 E, 12 P and 25 U**; each country has at least two E and at least one non-E case. They show observable diversity within a *purposefully selected panel*; the fractions 20/57, 12/57 and

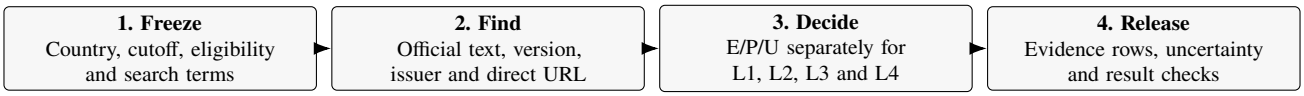


Fig. 1. Four-step *research workflow*. Arrows show audit steps, **not** causal transmission from national strategy to university policy. Black-and-white printing preserves all labels.

TABLE II
GCC RESULTS AT THE FIXED CUTOFF: FOUR DOCUMENTARY STATES AND THE INDEPENDENT 57-INSTITUTION CORROBORATION COUNTS. THE LAST THREE COLUMNS ARE *SELECTED-CASE COUNTS*, NOT NATIONAL PREVALENCE RATES.

Jurisdiction	L1	L2	L3	L4	E	P	U	How to interpret
United Arab Emirates	E	E	P	E	5	3	5	HE OBEF is a sector evaluation framework, not an L3 control under primary gate.
Saudi Arabia	E	E	E	E	3	4	4	Four gates have visible instruments; no transfer claim is made.
Qatar	E	E	P	E	2	2	6	University rules visible despite partial L3 evidence.
Oman	E	E	U	E	2	1	5	Recovered SQU guidance supports L4, not sector L3 [14].
Bahrain	P	E	U	E	6	0	1	School-oriented initiative does not qualify as HE-specific L3 [22].
Kuwait	P	P	U	E	2	2	4	Kuwait University rule provides an L4 example.
Total (selected HEIs)					**20**	**12**	**25**	**57 cases across six states**.

25/57 are *not representative population estimates*. The full 57-row links remain in the supplement instead of a micro-font table.

A. What changes when a source or rule changes?

TABLE III
EASY-TO-CHECK ALTERNATIVES: COUNTS, NOT CONFIDENCE INTERVALS

Declared scenario	L3 E	L4 E	D
Primary corpus	1/6	6/6	5/6
Without recovered Oman SQU source	1/6	5/6	4/6
Broader UAE OBEF gate	2/6	6/6	4/6
Exclude Saudi Arabia	0/5	5/5	5/5
Exclude any other one state	1/5	5/5	4/5

Table III states exactly which judgment is varied. If a new qualifying HE-sector source is recovered, the corresponding state must be recoded with its date and passage; it is *not* legitimate to treat a university policy as a ministry-level rule. As a purely algebraic check, if at most k of the six jurisdictional mismatch indicators in Eq. (1) change, the revised count satisfies $D' \geq \max(0, 5-k)$. This is a conditional threshold, *not* evidence that coding error is unlikely.

We retain the requested *interval-valued fuzzy representation* as a transparent sensitivity notation. For layer ℓ, if n_E states are E and n_P are P, the permissible envelope is

$$F_\ell = \left[\frac{n_E}{6}, \frac{n_E + n_P}{6}\right]. \tag{2}$$

For the observed counts, $F_3 = [1/6, 3/6]$ and $F_4 = [1, 1]$, so $F_4 - F_3$ has a minimum of $1/2$. This restates the *already known categorical* bounds; it is not an estimated probability, independently fitted fuzzy membership, missing-source correction, ranking method or uncertainty about coder error [20], [21].

VI. WHAT ANOTHER RESEARCHER CAN REUSE

The contribution is usable only if a reader can move from a conclusion back to the document that supports it. Every new audit should release: **(i)** a source register with issuer, edition, issue/effective date, language, document URL, exact clause and layer-specific decision; **(ii)** an eligibility list showing institutions included, excluded and tied at the boundary; and **(iii)** an E/P/U decision sheet with search/closure dates, explicit uncertainty flags and code that recalculates tables. An open hyperlink alone cannot substitute for the controlling provision or its date. A source URL that merely records an unsuccessful U search must be labelled as a *search locator*, not cited as a positive policy.

For university staff, we propose *eight public-rule questions*: who approved the operative version; whom it covers; what AI use is allowed in assessments; when disclosure is required; which tools or data are restricted; who verifies outputs; how fairness and appeals work; and who maintains future revisions. These fields address recurring traceability questions and align with general UNESCO/OECD governance concerns [23], [24]. They are a *documentation proposal*, not a validated policy-quality score. The companion field guide and editable worksheet turn each question into a source-linked check with blank evidence cells.

Supplementary materials and code. The **field guide and 57-institution linked appendix (PDF)** is available with the **version-matched research archive** (57-case CSV, six-state matrix, provisional 36-source register, blank audit template and

result checks). This is a public author-linked resource, not a blind-review artifact.

VII. Limits and Conclusion

The results concern *public document availability at a declared cutoff*. They do not measure instrument-to-instrument transmission, implementation effectiveness, policy quality, causation, regional prevalence or missing-source rates. The historical search log is incomplete for exact independent replay of every query; the source register is provisional until key quotations/editions and archival captures are independently checked. The 57-case membership remains subject to a full original-frame completeness check. The E/P/U states were not independently double-coded, so no inter-rater agreement or Arabic–English adjudication rate is claimed. A future verification round should freeze source copies, use two independent coders, report a confusion matrix and agreement, adjudicate conflicts with controlling passages and record any revised outcomes before publication. These are *outstanding validations*, not completed experiments.

The six-state result is simple: official university-level AI rules are publicly visible in every GCC jurisdiction, while the stricter HE-sector control gate is explicit in only one (two under the wider UAE interpretation). The portable finding is the *difference between visible evidence, explicit linkage and actual practice*, together with a reusable source-to-claim workflow. The open ledger and companion tools make the calculation auditable without presenting undocumented assumptions as established fact.

References

[1] M. R. Albous, O. R. Al-Jayyousi, and M. Stephens, "AI governance in the GCC states: A comparative analysis of national AI strategies," *Journal of Artificial Intelligence Research*, vol. 82, pp. 2389–2422, 2025, doi: 10.1613/JAIR.1.17619.

[2] Y. Jin, L. Yan, V. Echeverria, D. Gasevic, and R. Martinez-Maldonado, "Generative AI in higher education: A global perspective of institutional adoption policies and guidelines," *Computers and Education: Artificial Intelligence*, vol. 8, p. 100348, 2025, doi: 10.1016/j.caeai.2024.100348.

[3] N. McDonald *et al.*, "Generative artificial intelligence in higher education: Evidence from an analysis of institutional policies and guidelines," *Computers in Human Behavior: Artificial Humans*, vol. 3, p. 100121, 2025, doi: 10.1016/j.chbah.2025.100121.

[4] L. Parker *et al.*, "Comparative analysis of artificial intelligence policies in universities across five countries," *Discover Computing*, vol. 28, p. 267, 2025, doi: 10.1007/s10791-025-09745-5.

[5] B. L. Moorhouse, M. A. Yeo, and Y. Wan, "Generative AI tools and assessment: Guidelines of the world's top-ranking universities," *Computers and Education Open*, vol. 5, p. 100151, 2023.

[6] M. Li *et al.*, "A framework for developing university policies on generative AI governance: A cross-national comparative study," *Studies in Higher Education*, 2026, doi: 10.1080/03075079.2026.2696496.

[7] X. Jiang and Z. Abdullah, "AI-enabled governance in higher education: A systematic review of applications, outcomes, and emerging implications," *Frontiers in Education*, vol. 11, p. 1856440, 2026, doi: 10.3389/feduc.2026.1856440.

[8] M. S. Khoza, S. F. Wamba, and S. Nyawa, "Mapping the landscape of AI governance in higher education," *Assessment & Evaluation in Higher Education*, 2026, doi: 10.1080/02602938.2026.2712521.

[9] Y. An, J. H. Yu, and S. James, "Investigating higher education institutions' guidelines and policies regarding the use of generative AI in teaching, learning, research, and administration," *International Journal of Educational Technology in Higher Education*, vol. 22, p. 10, 2025.

[10] F. D. Bacalso *et al.*, "Institutional approaches to generative AI management in higher education: A systematic review," *Frontiers in Education*, vol. 11, p. 1814426, 2026, doi: 10.3389/feduc.2026.1814426.

[11] Y. Qian, "Governing generative AI in higher education: Emerging policy approaches and support ecosystems at innovative U.S. universities," *International Journal for Educational Integrity*, vol. 22, p. 25, 2026, doi: 10.1007/s40979-026-00233-x.

[12] M. Vukasovic, "Higher education governance and policy: An introduction to multi-issue, multi-level and multi-actor dynamics," *Policy and Society*, vol. 36, no. 1, pp. 1–15, 2017.

[13] D. Donina and S. Hasanefendic, "Higher education institutional governance reforms in the Netherlands, Portugal, and Italy: A policy translation perspective addressing the homogeneous/heterogeneous dilemma," *Higher Education Quarterly*, vol. 73, no. 1, pp. 29–44, 2019.

[14] Sultan Qaboos University, "Guidelines for using GenAI tools in teaching, learning and assessment," 2024. [Online]. Available: https://www.squ.edu.om/About/Support-Centers-/Centre-For-Excellence-In-Teaching-and-Learning.

[15] UAE Ministry of Higher Education and Scientific Research, "University guidebook for outcome-based evaluation framework (OBEF), version 11.5," 23 Mar. 2026. [Online]. Available: https://www.mohesr.gov.ae/Documents/OBF%20University%20Guide%20-%20EN.pdf.

[16] Education and Training Quality Authority (BQA), Bahrain, "Institutional review report: Kingdom University," 2019. [Online]. Available: https://www.bqa.gov.bh/En/Reports/UniReports/HigherEducationReport/KU%20Inst%20en%20C2.pdf.

[17] Education and Training Quality Authority (BQA), Bahrain, "Higher education reports," 2026; index modified 7 July 2026. [Online]. Available: https://www.bqa.gov.bh/institutions-performance/higher-education-reports/.

[18] Commission for Academic Accreditation, UAE, "Renewal of licensure: Emirates Institute for Banking and Financial Studies—executive summary," 2019. [Online]. Available: https://caa.ae/PORTALSUMMARYREPORTS/16/EIBFS%20Executive%20Summary.pdf.

[19] Commission for Academic Accreditation, UAE, "Higher education institution: Emirates Institute for Banking and Financial Studies," 2026. [Online]. Available: https://www.caa.ae/Pages/Institutes/Details.aspx?GUID=16.

[20] F. E. Petry and R. R. Yager, "Interval-valued fuzzy sets aggregation and evaluation approaches," *Applied Soft Computing*, vol. 124, p. 108887, 2022.

[21] B. Pękala, K. Dyczkowski, P. Grzegorzewski, and U. Bentkowska, "Inclusion and similarity measures for interval-valued fuzzy sets based on aggregation and uncertainty assessment," *Information Sciences*, vol. 547, pp. 1182–1200, 2021.

[22] Ministry of Education, Kingdom of Bahrain, "Digital empowerment in education program," updated 30 Dec. 2025; school-sector target groups. [Online]. Available: https://moe.gov.bh/en/digital-empowerment-in-education-program.

[23] UNESCO, *Guidance for Generative AI in Education and Research*. Paris: UNESCO, 2023.

[24] OECD, "Policies supporting responsible and systematic GenAI adoption in higher education," *OECD Education Spotlights*, no. 23, 2026, doi: 10.1787/c4e5621f-en.

[25] Qatar University, "Instructions on the use of generative artificial intelligence tools at Qatar University," approved/effective 25 Dec. 2025; English translation of Arabic original. [Online]. Available: https://www.qu.edu.qa/en-us/about/policy/Documents/Instructions%20onThe%20Use%20of%20Generative%20Artificial%20Intelligence%20Tools%20at%20Qatar%20University-Approved.pdf.